

Question	Answer	Marks
4(a)(i)	(vertically) upwards/up	B1
4(a)(ii)	increases (with time/velocity/depth)	B1
4(b)(i)	for a body in (rotational) equilibrium	B1
	<u>sum/total</u> of clockwise moments about a point = <u>sum/total</u> of anticlockwise moments about the (same) point	B1
4(b)(ii)	$(F_B \times 5.0)$ or (380×2.5) or (750×1.6)	C1
	$(F_B \times 5.0) = (380 \times 2.5) + (750 \times 1.6)$ $F_B = 430 \text{ N}$	A1
4(b)(iii)	taking moments about C: $(380 \times 2.5) = 750 \times (2.0 - x)$	C1
	$(2.0 - x) = 1.3$ $x = 0.7 \text{ m}$	A1
	or	
	moments may be taken about other points, e.g. about D: $(380 \times 4.5) + (750 \times x) = 1130 \times 2.0$	(C1)
	$x = 0.7 \text{ m}$	(A1)